Timestamp formats, ordering and expiry rules

The rules are tied to repository contracts, not invented deadlines. Re-evaluate freshness at the time of the actual action; this report does not extend validity. No approvals or event times were created by the audit.

Context	Required rule	Source
Format	Instants use YYYY-MM-DDTHH:mm:ss.sssZ or an explicit UTC offset. Preserve YYYY-MM-DD for a business date. UTC is canonical; Paris/Douala columns are display conversions.	Extraction database type metadata
Row creation/update	createdAt/updatedAt describe the stored row. They cannot substitute for appointment, role assignment, qualification, review or approval timestamps.	prisma/schema.prisma; _UserRoles columns A/B only
Common sensitive actions	The audited role-member changes, AP confirmation, journal posting, close certification, branch signing, payroll review/approve/post and AI approval/activation paths require server-verified fresh authentication within 300 seconds. Each exact action remains authoritative.	Prior capability-contracts.json source anchors; services/purchasing/payment-request-authority.ts:9
Module composition	Fresh authentication within 600 seconds, plus current revision/preview hash and entitlement prerequisites.	Prior CAP-MODULE-001 source anchors
POS G1 approval	freshAuthenticatedAt <= approvedAt <= validation time; approvedAt - freshAuthenticatedAt <= 600000 ms. Decision reviewOrExpiryAt must be after validation time.	scripts/pos-g1-contract-gate.js:136
POS evidence/trust key	approvedAt <= proof.verifiedAt <= now; proof.reviewOrExpiryAt > now and > proof.verifiedAt; proof expiry <= decision expiry and key.validUntil; proof.verifiedAt >= key.validFrom; now < key.validUntil.	scripts/pos-g1-contract-gate.js:143; scripts/pos-governance-evidence.js:81
Country review	Use actual requiredApproval.reviewedAt and source-backed effectiveFrom/effectiveTo. The current static gate only checks nonempty reviewedAt/effectiveFrom; effectiveTo is not enforced. Do not treat that weak check as qualification/date verification.	scripts/statutory-country-pack-production-gate.js; CM_CNPS_PRODUCTION_APPROVAL contract
Accountant consent	effectiveFrom <= now < expiresAt, with ACTIVE status and exact target-client/accountant scope.	services/accounting/accountant-access.service.ts; CAP-ACCOUNTANT-001
Agent approval/coverage	decidedAt and acceptedAt are actual past/current events; expiresAt and coverageEndsAt must remain future. Coverage starts no later than the pilot and ends no earlier than pilot end; optional validUntil must remain current.	scripts/agent-external-input-readiness.js:284; services/agents/agent-release-control.service.ts:820
Agent certification	passedAt and expiresAt accompany the genuine CI report; expiry must follow passedAt and remain current, bound to the exact commit/manifest.	services/agents/agent-release-control.service.ts:460,854
Release/CI evidence	CI completedAt and attestedAt, governance attestedAt, deployment attestation and credential authority attestedAt must be no older than 24 hours and not future. CI completion <= attestation; deployment <= deployment attestation.	scripts/agent-operational-release-gate.js:235,309,566; scripts/agent-credential-rotation-gate.js:312

Context	Required rule	Source
Scheduler	intervalMinutes=5; 0<timeoutMs<300000. At least three unique completed windows; scheduledAt <= completedAt <= now. Consecutive schedules: five minutes +/-30 seconds. readinessCheckedAt no older than ten minutes; heartbeatFreshUntil > now.	scripts/agent-operational-release-gate.js:532,594,625
Alert evidence	deliveredAt no older than 24 hours; deliveredAt <= acknowledgedAt <= now; acknowledgement within configured integer SLO of 1..120 minutes. escalationTestedAt is an actual past/current test.	scripts/agent-operational-release-gate.js:691
Credential rotation	For ROTATED_AND_REVOKED: rotationStartedAt <= newVersionActivatedAt <= dependentWorkloadsRestartedAt <= newVersionVerifiedAt <= oldVersionRevokedAt <= oldVersionRejectedAt <= reviewedAt. Completed classification also requires authentic reviewedAt and authority binding.	scripts/agent-credential-rotation-gate.js:257,392
No activation yet	Null activatedAt is appropriate when activation has not occurred. Never populate it merely to clear an audit. Revocation/deletion/termination dates similarly remain null unless the event applies.	Existing pending-release templates and guarded runtime workflows
Business effect versus observation	effectiveAt/hireDate/periodStart/source publication dates represent business context. observedAt records when this audit read the data. A date in a seed row can be synthetic even if well formatted.	Current database extraction and source provenance